A Consolidated Adelic Spectral-Flow Program for the Riemann Hypothesis

Merged Reviewer Edition: original paper plus computational appendix, visual supplement, and non-specialist reader guide

Purpose of this merged edition

This single PDF is intended for a reviewer who sees only one document. It preserves the original paper, adds a plain-language reader guide before it, and then appends the extended computational appendix with the first 100 primes and the visualization suite. The front matter is explanatory; it is not a claim of a proof.

How to read this document

The document has three parts. Part I is this reader guide. Part II is the original research program, which frames a conditional route toward the Riemann Hypothesis using Li coefficients, Weil decompositions, b-calculus, APS boundary theory, and gamma-compressed positivity. Part III is the computational appendix, which clarifies the role of primes, verifies the first 100 primes by an elementary algorithm, and adds visual material.

Part	What it contains	Recommended use
I. Reader guide	Plain-language map, definitions, and caveats.	Read first if the research area is unfamiliar.
II. Original paper	The consolidated spectral-flow program and theorem stack.	Read for the mathematical proposal and conditional claims.
III. Appendix	Prime verification method, first 100 primes, figures, and toy visualizations.	Use to check computations and understand the prime-side notation.

Critical caveat

Neither the original paper nor the appendix proves the Riemann Hypothesis. The original paper describes a conditional program and isolates obstructions. The appendix adds reproducible computations and visual explanation, but it does not turn the conditional program into a proof.

Plain-language overview

The Riemann Hypothesis concerns the non-trivial zeros of the zeta function. A classical equivalent formulation, Li's criterion, replaces the location of the zeros with the positivity of a sequence of real numbers called the Li coefficients. The original paper tries to reinterpret these Li coefficients using geometric and spectral language: Dirac-type operators, boundary conditions, spectral flow, and compressed kernels.

The central schematic idea is that each Li coefficient should be expressible in a form resembling

$$\lambda_n = I_n - (1/2)\, h_n,$$

where I_n is a renormalized boundary or spectral-flow invariant, and h_n is a zero-mode obstruction. If one could prove both $h_n = 0$ and $I_n \geq 0$ for every n, then $\lambda_n \geq 0$ for every n, and Li's criterion would imply RH.

The source paper is careful about what remains unproved. It emphasizes that the ordinary integer-valued index language cannot literally represent real-valued Li coefficients or Archimedean terms. It also isolates a substantial obstruction: subtracting a scalar Archimedean trace defect is not enough to prove operator positivity on the compressed gamma-Sonine subspace.

Key terms in one page

Li coefficients	A sequence λ_n whose nonnegativity for all n is equivalent to RH.
Weil decomposition	A split of λ_n into a prime-orbit contribution B_n and an Archimedean contribution A_n.
Prime side B_n	A weighted sum over primes. It uses primes as inputs; it does not generate primes.
Archimedean term A_n	A real-valued contribution involving the gamma factor. In the paper it is the negative term that must be controlled.
APS boundary theory	A framework for relating operators on spaces with boundary to spectral data at the boundary.
Zero mode	A kernel element of the deformed operator D_n. The program needs these to vanish.
Gamma compression	A proposed way to isolate or remove the Archimedean gamma contribution in a boundary model.
Domination inequality	The non-circular positivity target needed to keep negative spectral directions from surviving compression.

What the appendix adds

The appendix was added because a reviewer asked whether the paper contained a formula for generating prime numbers. The answer is no. The original paper contains formulas where primes appear inside sums, especially the prime-orbit term. Those formulas require primes as an indexing set; they do not output primes.

To make that distinction concrete, the appendix supplies a transparent deterministic procedure for generating and verifying the first 100 primes: trial division by previously found primes up to the square root of each candidate. It then lists the 100 primes, their gaps, the number of trial divisions used for each accepted prime, and a collection of visualizations. These computations are elementary but useful because they anchor the prime-side notation in explicit data.

What was verified computationally

The appendix verifies the first 100 primes and computes simple derived statistics: prime gaps, cumulative sums, reciprocal sums, and prime-side local weights such as log(p)/sqrt(p) and log(p)/p. These are sanity checks and explanatory visuals, not RH-level evidence.

What was not verified computationally

The appendix does not compute the actual Li coefficients from the paper, because the uploaded synthesis does not provide explicit computable probe functions f_n. It also does not verify the APS-Li formula, the zero-mode theorem, the boundary positivity conjecture, or the gamma-compressed domination inequality.

Reviewer checklist

• Separate definitions, conjectures, conditional theorems, and proved propositions.

• Check whether each positivity statement is scalar, trace-level, or operator-level.

• Do not read the prime-side sum as a prime generator.

• Treat toy Gaussian log-probe figures as explanatory diagnostics only.

• Use the appendix for reproducibility of the first-100-prime verification, not as evidence that RH has been proved.

Map of the merged PDF

The original paper begins immediately after this guide. It is followed by the appendix. The sections below are the most important landmarks for a first pass.

Original paper, Sections 2-5	Li coefficients, Weil decomposition, and minimal theorem stack.
Original paper, Section 6	Zero-mode problem and b-Weitzenboeck coercivity.
Original paper, Sections 7-10	Boundary positivity, gamma-weighted Sonine compression, compactness, and eta correction.
Original paper, Sections 11-12	Gamma-compressed kernel obstruction and corrected zero-mode implication.
Original paper, Sections 14-15	Falsifiable research program and final conditional synthesis.
Appendix, Sections 4-5	Deterministic verification method and table of the first 100 primes.
Appendix, Sections 6-8	Prime-side local weights, visualizations, and toy prime-side sums.

One-sentence summary

This merged edition presents a conditional RH research program and a reproducible computational supplement; its main value is clarification, falsifiability, and organization, not a completed proof.

Part II

Merged reviewer edition

Original Research Program

The uploaded source paper is reproduced next.
It presents the spectral-flow framework, theorem stack, and open obstructions.

Reader note

Read this as the mathematical proposal. It is conditional and explicitly does not claim a proof of the Riemann Hypothesis.

A Consolidated Adelic Spectral-Flow Program for the Riemann Hypothesis

Li Positivity, Sonine Compression, b-Calculus, Witt Gauge Geometry, the Renormalized Boundary Defect, and the Gamma-Compressed Kernel Obstruction

Synthesized from the supplied manuscripts and subsequent analysis

Attribution: original program authors and collaborative AI-review extensions;
this extension adds the gamma-compressed Weil-kernel positivity analysis and negative-support obstruction.

April 2026 synthesis; extended version

Abstract

This paper gives a self-contained synthesis of a sequence of supplied manuscripts developing a proposed program toward the Riemann Hypothesis. The program attempts to unify Connes' noncommutative geometry of the adele-class space, the Berry–Keating $H = xp$ picture, $\mathbb{F}_1$-geometric positivity, Sonine-space compression, and supergeometric spectral flow through Atiyah–Patodi–Singer (APS) boundary theory. The present article does not claim a proof of the Riemann Hypothesis. Instead it isolates the strongest coherent theorem stack that emerges from the program and makes explicit where the remaining RH-level difficulty resides.

The main consolidation is this: Li positivity may be attacked through an APS/spectral-flow formula of the schematic form

$$\lambda_n = I_n - \frac{1}{2}h_n,$$

where $h_n = \dim\ker D_n$ is the zero-mode obstruction and I_n is a boundary-renormalized spectral invariant. The zero-mode problem is reduced to a b-Weitzenboeck coercivity inequality. The positivity of the boundary invariant is reformulated through a gamma-weighted Sonine compression and a Schur/Herglotz boundary transfer function. A b-regularized compactness theorem is proposed for the gamma-Sonine boundary projection. A crucial correction is also made: the ordinary Fredholm pair index is integer-valued and therefore cannot literally equal the real Archimedean term in the Weil formula. The appropriate object is a renormalized relative eta or spectral-shift defect.

This extended version adds a further obstruction theorem. The gamma-compressed Weil kernel is positive definite only if the gamma-Sonine projection removes all negative spectral directions of the Mellin multiplier, not merely the scalar Archimedean trace defect. Finite gamma-moment compression cannot generally achieve this when the negative set has positive measure. Thus the next exact target is a strong spectral-support theorem, $P_\Gamma \le 1_{\{m_n \ge 0\}}(M_{m_n})$, or an equivalent domination inequality on the gamma-Sonine subspace.

Contents

1 Attribution, scope, and status **3**

1.1 The central conclusion 3

2 Classical input: Li coefficients and the Weil decomposition **4**

2.1 Li's criterion 4

2.2 The Weil decomposition 4

3 The unified geometric architecture **4**
3.1 Resolution I: Sonine compression . . . 5
3.2 Resolution II: graded Clifford-adeles and spectral flow . . . 5

4 What the five manuscripts contributed **5**
4.1 Original Vision Paper . . . 5
4.2 Revision 1 / Version 3 . . . 6
4.3 Revision 2 / Version 5 . . . 6
4.4 Quantum proposals, Versions 6.3 and 6.4 . . . 6

5 The minimal theorem stack **6**

6 The zero-mode problem via b-Weitzenboeck coercivity **7**
6.1 Li–Witt convexity and the Riccati route . . . 8

7 Boundary positivity and the Schur/Herglotz route **9**
7.1 The Li generating series and disk transfer functions . . . 9
7.2 Why ordinary Sonine compression is insufficient . . . 9
7.3 Gamma-weighted Sonine compression . . . 9

8 Compactness of the b-regularized gamma-Sonine projection **10**
8.1 Boundary model . . . 10
8.2 Renormalized gamma defect . . . 10

9 The index mismatch and the renormalized eta correction **11**
9.1 Renormalized relative eta defect . . . 11

10 Revised statement of Lemma B **12**

11 The gamma-compressed Weil kernel and the new obstruction **12**
11.1 Mellin multiplier form . . . 13
11.2 Finite gamma compression cannot generally prove positivity . . . 13
11.3 Negative-support exclusion is insufficient by itself . . . 14
11.4 The exact revised target . . . 14

12 Consequences for the zero-mode route **14**

13 The quantum branch as a dependent application **15**

14 Falsifiable research program **15**
14.1 Finite-dimensional truth laboratory . . . 15
14.2 Analytic tasks . . . 16

15 Final consolidated theorem **16**

16 Conclusion **16**

1 Attribution, scope, and status

This paper consolidates and extends five supplied manuscripts:

(i) *A Vision Paper: A Grand Unified Program for the Riemann Hypothesis via $\mathbb{F}_1$ Geometry, Berry–Keating Scaling, and Supergeometric Spectral Flow*, here cited as the Original Vision Paper.

(ii) *A Grand Unified Program for the Riemann Hypothesis, Version 3: Advances on All Ten Research Tasks*, here cited as Revision 1 or Version 3.

(iii) *A Grand Unified Program for the Riemann Hypothesis, Version 5: b-Calculus and Weitzenboeck Geometry*, here cited as Revision 2 or Version 5.

(iv) *A Grand Unified Program for the Riemann Hypothesis, Version 6.3: Effective Adelic Spectral Flow, Quantum Register Embedding, and an Intrinsically Fault-Tolerant Topological Factoring Oracle*, here cited as Proposal Foundations.

(v) *A Grand Unified Program for the Riemann Hypothesis, Version 6.4: Effective Adelic Spectral Flow, Quantum Register Embedding, and an Intrinsically Fault-Tolerant Topological Factoring Oracle*, here cited as Quantum Proposal.

The present synthesis also incorporates subsequent analysis: the "dream theorem" reduction to b-Weitzenboeck coercivity, the gamma-weighted Sonine projection, the Schur/Herglotz boundary reformulation, the compactness route for the b-regularized gamma correction, the correction distinguishing an integer Fredholm index from a real renormalized eta defect, and the newly isolated obstruction to positive definiteness of the gamma-compressed Weil kernel.

Remark 1.1 (No proof claim). No proof of the Riemann Hypothesis is claimed. Several statements below are conditional. The paper's purpose is to form one cohesive and rigorous research program, state the best available conditional theorem stack, and isolate the exact points where a proof would still need to be supplied.

1.1 The central conclusion

The best consolidated program is not the quantum-factoring branch. The strongest mathematical core is

$$\text{Li positivity} \quad \Longleftarrow \quad \text{APS boundary positivity + zero-mode vanishing.}$$

More explicitly, the program seeks a formula of the form

$$\lambda_n = I_n - \frac{1}{2}h_n, \qquad h_n = \dim\ker(D_n), \tag{1}$$

where λ_n are the Li coefficients, $D_n = D_0 + A(f_n)$ is a gauge-deformed Dirac-type operator, and I_n is the relevant APS/spectral-flow invariant. If

$$h_n = 0, \qquad I_n \geq 0$$

for every $n \geq 1$, then Li's criterion gives RH.

The essential work splits into two tasks:

Task 1. Prove the zero-mode theorem $\ker D_n = 0$.

Task 2. Prove positivity of the correct boundary invariant I_n.

The rest of this paper develops the most precise known formulation of these two tasks.

2 Classical input: Li coefficients and the Weil decomposition

2.1 Li's criterion

Let

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$$

be the completed zeta function. Li's coefficients are

$$\lambda_n = \sum_{\rho}\left[1 - \left(1 - \frac{1}{\rho}\right)^n\right], \tag{2}$$

where the sum ranges over non-trivial zeros of $\zeta(s)$ in the usual symmetric sense. Equivalently,

$$\lambda_n = \frac{1}{(n-1)!}\frac{d^n}{ds^n}\left[s^{n-1}\log\xi(s)\right]_{s=1}. \tag{3}$$

Li's theorem states

$$RH \quad \Longleftrightarrow \quad \lambda_n \geq 0 \quad \text{for all } n \geq 1. \tag{4}$$

This is the arithmetic target of the whole program.

2.2 The Weil decomposition

The Original Vision Paper emphasizes that the explicit formula gives, for normalized Li probes f_n,

$$\lambda_n = A_n + B_n + C_n. \tag{5}$$

In the normalized case $C_n = 0$, so

$$\lambda_n = A_n + B_n. \tag{6}$$

Here B_n is the positive prime-orbit contribution and A_n is the negative Archimedean contribution. In schematic form,

$$B_n = \sum_p \sum_{m\geq 1} \frac{\log p}{p^{m/2}}\left(f_n(p^m) + f_n(p^{-m})\right) > 0, \tag{7}$$

while

$$A_n = -\frac{1}{2\pi}\int_{-\infty}^{\infty} f_n\left(\frac{1}{2}+it\right)\frac{\Gamma'}{\Gamma}\left(\frac{1}{4}+\frac{it}{2}\right)dt < 0. \tag{8}$$

Thus RH is not equivalent to mere positivity of the prime side. It requires the domination

$$B_n \geq |A_n| \qquad \forall n \geq 1. \tag{9}$$

This is the basic obstruction around which every subsequent construction must be organized.

3 The unified geometric architecture

The Original Vision Paper identifies three major lines of thought and proposes that they are different faces of the same structure:

1. Connes' noncommutative geometry on the adele-class space $X = \mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^{\times}$.
2. The Berry–Keating quantization of the Hamiltonian $H = xp$.
3. $\mathbb{F}_1$-geometric intersection positivity.

The proposed bridge is the multiplicative scaling action on the adele-class space. If $u \in \mathbb{R}_+^\times$, the scaling operator on a suitable Hilbert space $\mathcal{H} = L^2(X, d^*x)$ is

$$U(f)\psi(x) = \int_{\mathbb{R}_+^\times} f(u)\psi(xu)\frac{du}{u}. \tag{10}$$

Its infinitesimal generator is the Berry–Keating operator

$$D = x\frac{d}{dx} + \frac{1}{2}. \tag{11}$$

The regularized trace of $U(f_n)$ recovers the Weil distribution:

$$\mathrm{Tr}_{\mathrm{reg}}\, U(f_n) = \lambda_n. \tag{12}$$

But because $A_n < 0$, $U(f_n)$ cannot be globally positive on the full Hilbert space. The program therefore proposes two complementary repairs.

3.1 Resolution I: Sonine compression

Let P_{Son} be a projection onto a Sonine-type subspace of Mellin test functions. The original conjecture is that the compressed operator is positive:

$$P_{\mathrm{Son}} U(f_n) P_{\mathrm{Son}} \geq 0. \tag{13}$$

This is meant to remove or control the negative Archimedean scattering contribution. It is a falsifiable operator inequality: finite truncations should reveal any persistent negative eigenvalues.

3.2 Resolution II: graded Clifford-adeles and spectral flow

The second mechanism introduces a $\mathbb{Z}_2$-graded Hilbert module

$$\mathcal{H} = \mathcal{H}_+ \oplus \mathcal{H}_-, \tag{14}$$

where the positive component models prime orbits and the negative component models Archimedean scattering. The ordinary trace is replaced by a supertrace:

$$\mathrm{STr}(U(f_n)) = \mathrm{Tr}_{\mathcal{H}_+}(U(f_n)) - \mathrm{Tr}_{\mathcal{H}_-}(U(f_n)) = B_n - |A_n| = \lambda_n. \tag{15}$$

A Clifford-valued Dirac operator D_0 is then deformed by a Witt-vector gauge potential:

$$D_n = D_0 + A(f_n). \tag{16}$$

RH is recast as a positivity or monotonicity statement for the spectral flow of the family $D_t = D_0 + tA(f_n)$.

4 What the five manuscripts contributed

4.1 Original Vision Paper

The Original Vision Paper is conceptually strongest where it is cautious. It states no proof claim, identifies the Archimedean term as the obstruction, and proposes two falsifiable paths: Sonine compression and supergeometric spectral flow. It also identifies a McKean–Singer obstruction to a static factorization and therefore motivates a dynamic spectral-flow formulation.

4.2 Revision 1 / Version 3

Revision 1 adds three important components:

1. a $\mathbb{Z}_2$-grading regularity statement;
2. the vanishing of the eta invariant $\eta(D_0) = 0$ by spectral symmetry;
3. an APS formula of the schematic form

$$\lambda_n = \mathrm{ind}_{\mathrm{APS}}(D_0, P_n) - \frac{1}{2}h_n, \qquad h_n = \dim\ker(D_n). \tag{17}$$

This identifies $h_n = 0$ as the decisive obstruction. Revision 1 also reports positivity of computed Li coefficients up to $n = 100$, but those numerics are sanity checks rather than evidence at the level needed for RH.

4.3 Revision 2 / Version 5

Revision 2 is the strongest analytic revision. It introduces b-calculus and weighted Sobolev spaces to handle the noncompact adele-class space. The Archimedean component is treated as a cylindrical end, and the Fredholm property is restored by b-ellipticity and an invertible indicial family. It also develops a Weitzenboeck approach to $h_n = 0$ and states a generating-series reformulation:

$$G(z) = \sum_{n\geq 1} \lambda_n z^n. \tag{18}$$

The proposed singularities are

$$z_\rho = \frac{\rho}{\rho - 1}, \tag{19}$$

with $|z_\rho| = 1$ if and only if $\mathrm{Re}(\rho) = 1/2$. Thus RH is equivalent to analyticity of G in $|z| < 1$, subject to the necessary convergence and analytic-continuation controls.

4.4 Quantum proposals, Versions 6.3 and 6.4

The quantum branch proposes an $O(\log N)$-qubit embedding of a localized Bost–Connes module, adiabatic deformation by a Witt gauge potential, quantum phase estimation of an APS index, and a topologically protected factoring oracle. Its conceptual contribution is the idea that a topological invariant may be robust to local noise.

However, this branch remains speculative for two reasons. First, the polynomial-time claim depends on a conjectural adiabatic spectral gap. Second, the readout appears to require efficient localization on prime-factor support without already knowing the factors. Therefore, the quantum material should be treated as a possible application of a future analytic theorem, not as the foundation of the RH program.

5 The minimal theorem stack

The consolidated program should now be stated as a theorem stack.

Conjecture 5.1 (APS-Li formula)**.** *For each Li probe f_n there is a b-Fredholm Dirac-type operator D_0, an admissible APS boundary projection P_n, and a gauge-deformed operator $D_n = D_0 + A(f_n)$ such that*

$$\lambda_n = I_n - \frac{1}{2}h_n, \qquad h_n = \dim\ker(D_n), \tag{20}$$

where I_n is the appropriate renormalized APS/spectral-flow invariant.

Conjecture 5.2 (Zero-mode theorem)**.** *For every $n \geq 1$,*

$$\ker(D_n) = \{0\}. \tag{21}$$

Conjecture 5.3 (Boundary positivity)**.** *For every $n \geq 1$,*

$$I_n \geq 0. \tag{22}$$

Theorem 5.4 (Conditional RH theorem)**.** *Assume Conjectures 5.1, 5.2, and 5.3. Then RH holds.*

Proof. By Conjecture 5.2, $h_n = 0$ for all n. By Conjecture 5.1, $\lambda_n = I_n$. By Conjecture 5.3, $\lambda_n \geq 0$ for all n. Li's criterion gives RH. □

6 The zero-mode problem via b-Weitzenboeck coercivity

The zero-mode obstruction is the most concrete part of the program. On the p-sector of the Bost–Connes–Clifford module and in the b-cylinder coordinate $u = \log x$, take the local normal form

$$D_{n,p} = \gamma^1 \log p + \gamma^2 \left(\partial_u + \frac{1}{2} + a_{n,p}(u)\right), \tag{23}$$

with Clifford relations

$$(\gamma^1)^2 = (\gamma^2)^2 = 1, \qquad \gamma^1\gamma^2 + \gamma^2\gamma^1 = 0. \tag{24}$$

The scalar $a_{n,p}(u)$ is a local representative of the Witt gauge potential. A natural regularized representative is

$$a_{n,p}(u) = (\log p)\chi(u) \operatorname{Re} f_n(pe^u), \tag{25}$$

where $\chi \in C_c^\infty(\mathbb{R})$ is a b-cylinder cutoff.

Proposition 6.1 (Local b-Weitzenboeck formula)**.** *For $D_{n,p}$ as in (23),*

$$D_{n,p}^* D_{n,p} = -\partial_u^2 + (\log p)^2 + \left(\frac{1}{2} + a_{n,p}(u)\right)^2 - a_{n,p}'(u). \tag{26}$$

Proof. The formal adjoint in the b-measure is

$$D_{n,p}^* = \gamma^1 \log p + \gamma^2 \left(-\partial_u + \frac{1}{2} + a_{n,p}(u)\right).$$

The mixed Clifford terms cancel by anticommutation. Writing $b(u) = 1/2 + a_{n,p}(u)$,

$$(-\partial_u + b)(\partial_u + b) = -\partial_u^2 + b^2 - b'.$$

This gives (26). □

Theorem 6.2 (Conditional zero-mode theorem)**.** *Suppose that, for every n, there exists $\kappa_n > 0$ such that*

$$(\log p)^2 + \left(\frac{1}{2} + a_{n,p}(u)\right)^2 - a_{n,p}'(u) \geq \kappa_n \tag{27}$$

for all relevant primes p and all u in the b-cylinder model. Then $\ker(D_n) = \{0\}$.

Proof. The global b-Weitzenboeck formula has the form

$$D_n^* D_n = (\nabla^{(n)})^* \nabla^{(n)} + V_n,$$

with V_n locally given by the scalar potential in (27). The APS boundary contribution is nonnegative because the APS projection removes the negative spectral modes of the boundary operator. Therefore, for ψ in the APS/b-domain,

$$\|D_n\psi\|^2 \geq \|\nabla^{(n)}\psi\|^2 + \kappa_n\|\psi\|^2 \geq \kappa_n\|\psi\|^2.$$

If $D_n\psi = 0$, then $\psi = 0$. □

Since $p \geq 2$, a sufficient form of (27) is

$$\sup_{p,u}\left[a'_{n,p}(u) - \left(\frac{1}{2} + a_{n,p}(u)\right)^2\right] < (\log 2)^2. \tag{28}$$

This is the Li–Witt coercivity inequality. It is the most concrete formulation of the zero-mode problem currently available.

6.1 Li–Witt convexity and the Riccati route

Let

$$b_{n,p}(u) = \frac{1}{2} + a_{n,p}(u).$$

The sharper Riccati inequality

$$b'_{n,p}(u) \leq b_{n,p}(u)^2 \tag{29}$$

implies (28), since then the potential in (26) is at least $(\log p)^2 \geq (\log 2)^2$.

A useful sufficient mechanism is an outer-amplitude representation. Suppose

$$b_{n,p}(u) = -\frac{d}{du}\log y_{n,p}(u), \qquad y_{n,p}(u) > 0. \tag{30}$$

Then

$$b'_{n,p}(u) = b_{n,p}(u)^2 - \frac{y''_{n,p}(u)}{y_{n,p}(u)}. \tag{31}$$

Thus $y''_{n,p}(u) \geq 0$ implies the Riccati inequality.

Proposition 6.3 (Positive spectral measure implies Li–Witt convexity)**.** *If the gamma-compressed prime-sector amplitude has the Laplace representation*

$$y_{n,p}(u) = \int_0^\infty e^{-u\xi}\, d\mu_{n,p}(\xi), \qquad \mu_{n,p} \geq 0, \tag{32}$$

then $y''_{n,p}(u) \geq 0$. *Consequently* (29) *holds and* $D_n^* D_n \geq (\log 2)^2 I$.

Proof. Differentiating under the integral gives

$$y''_{n,p}(u) = \int_0^\infty \xi^2 e^{-u\xi}\, d\mu_{n,p}(\xi) \geq 0.$$

Then (30) gives (29); Proposition 6.1 gives the lower bound. □

Thus the zero-mode problem is converted into a positive-measure problem for the gamma-compressed prime-sector amplitude. The subsequent sections analyze whether the gamma compression really supplies such a positive measure.

7 Boundary positivity and the Schur/Herglotz route

The boundary positivity conjecture should not be phrased as positivity of an ordinary integer APS index. The Li coefficients and Archimedean terms are real-valued. Therefore, the correct object must be a renormalized APS invariant: a relative eta, a spectral-shift functional, or a renormalized supertrace.

7.1 The Li generating series and disk transfer functions

Revision 2 introduced the generating function

$$G(z) = \sum_{n\geq 1} \lambda_n z^n.$$

A more structural disk object is

$$F(z) = \xi\left(\frac{1}{1-z}\right). \tag{33}$$

Then

$$\frac{d}{dz}\log F(z) = \sum_{n\geq 1} \lambda_n z^{n-1}, \qquad z\frac{F'(z)}{F(z)} = G(z). \tag{34}$$

Thus the Li generating series is a logarithmic derivative of a disk-transformed completed zeta function.

The Schur/Herglotz route asks whether the associated boundary transfer function is contractive in the unit disk. A generic Schur function Θ is characterized by positivity of the de Branges–Rovnyak kernel

$$K_\Theta(z,w) = \frac{1-\Theta(z)\overline{\Theta(w)}}{1-z\overline{w}}. \tag{35}$$

If the Li boundary transfer function is Schur, the corresponding Toeplitz/relative boundary object has a positive orientation. This is the analytic form of APS boundary positivity.

7.2 Why ordinary Sonine compression is insufficient

The original Sonine condition annihilates Mellin data near the critical line, but the Archimedean contribution (8) involves the entire gamma kernel

$$\frac{\Gamma'}{\Gamma}\left(\frac{1}{4}+\frac{it}{2}\right).$$

Therefore, ordinary Sonine compression does not obviously remove the full negative Archimedean term.

7.3 Gamma-weighted Sonine compression

Define the Archimedean gamma functional

$$\mathcal{A}_\Gamma(\varphi) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \varphi(t)\frac{\Gamma'}{\Gamma}\left(\frac{1}{4}+\frac{it}{2}\right)dt. \tag{36}$$

The gamma-weighted Sonine space is the closed subspace

$$\mathcal{S}_\Gamma = \{\varphi : \mathcal{A}_\Gamma(\varphi) = 0\}, \tag{37}$$

or, in a finite-mode model, the common kernel of finitely many regularized gamma moment functionals. Let P_Γ be the projection onto this space. The intended effect is

$$P_\Gamma(A\text{-term})P_\Gamma = 0, \qquad P_\Gamma U(f_n)P_\Gamma \sim \text{positive prime contribution.} \tag{38}$$

This is the correct target for Lemma B: it attacks exactly the negative term rather than a vague neighborhood of the critical line.

8 Compactness of the b-regularized gamma-Sonine projection

We now state the strongest rigorous bridge between the Li APS boundary condition P_n and the gamma-weighted Sonine projection P_Γ.

8.1 Boundary model

Let

$$\mathcal{H}_\partial = L^2(\mathbb{R}_t; \mathcal{K}) \tag{39}$$

be the Mellin-side boundary Hilbert space, where t is the Fourier-Mellin variable of the Archimedean cylinder and $\mathcal{K}$ is the prime-Clifford multiplicity space. Let B_n be the Li boundary operator and

$$P_n = 1_{[0,\infty)}(B_n). \tag{40}$$

The raw gamma multiplier

$$M_\psi(t) = \frac{\Gamma'}{\Gamma}\left(\frac{1}{4} + \frac{it}{2}\right) \tag{41}$$

is not compact. Indeed, as $|t| \to \infty$,

$$\frac{\Gamma'}{\Gamma}\left(\frac{1}{4} + \frac{it}{2}\right) = \log|t| + O(|t|^{-1}). \tag{42}$$

Multiplication by $\log|t|$ on $L^2(\mathbb{R})$ is not compact. Therefore the full gamma multiplier cannot be used as a compact perturbation.

8.2 Renormalized gamma defect

Separate the principal asymptotic part and retain only the short-range defect:

$$m_\Gamma(t) = \frac{\Gamma'}{\Gamma}\left(\frac{1}{4} + \frac{it}{2}\right) - \log(1 + |t|). \tag{43}$$

Then $m_\Gamma(t) = O(|t|^{-1})$ at infinity, modulo lower-order regularization near $t = 0$. Let $\chi \in C_c^\infty(\mathbb{R})$ be a spectral cutoff. Define

$$R_\Gamma = \chi(B_n) M_{m_\Gamma} \chi(B_n), \tag{44}$$

and

$$P_\Gamma = 1_{[0,\infty)}(B_n + R_\Gamma). \tag{45}$$

Theorem 8.1 (Compactness of the b-regularized gamma correction)**.** *In the b-regularized boundary model above,*

$$R_\Gamma \in \mathcal{K}(\mathcal{H}_\partial). \tag{46}$$

Proof. The operator $\chi(B_n)$ is spectrally compactly supported and smoothing in the b-calculus. In the exact cylinder model, it is a Fourier multiplier with smooth compact support; in the geometric b-model it has a smooth b-kernel with compactly supported normal operator. The multiplier M_{m_Γ} is bounded on the support selected by χ. Hence $\chi(B_n)M_{m_\Gamma}\chi(B_n)$ is Hilbert–Schmidt in the cylinder model and a compact residual b-operator in the geometric model. Thus R_Γ is compact. □

Theorem 8.2 (Compactness of the projection difference)**.** *Assume the b-weight is chosen so that B_n has an essential spectral gap around zero and the path $B_t = B_n + tR_\Gamma$ has no essential-spectrum crossing at zero. Then*

$$P_\Gamma - P_n \in \mathcal{K}(\mathcal{H}_\partial). \tag{47}$$

Proof. For $z \notin \mathbb{R}$, the resolvent identity gives

$$(B_n + R_\Gamma - z)^{-1} - (B_n - z)^{-1} = -(B_n + R_\Gamma - z)^{-1} R_\Gamma (B_n - z)^{-1}.$$

Since R_Γ is compact and the resolvents are bounded, the resolvent difference is compact. Choose a smooth function f equal to 0 below $-\varepsilon$ and 1 above ε, where $(-\varepsilon, \varepsilon)$ misses the essential spectrum. By the Helffer–Sjoestrand functional calculus, $f(B_n + R_\Gamma) - f(B_n)$ is compact. The difference between $f(B)$ and the sharp projection $1_{[0,\infty)}(B)$ occurs only in the finite-dimensional discrete spectral window. Therefore $P_\Gamma - P_n$ is compact. □

Corollary 8.3 (Fredholm pair)**.** *The pair (P_Γ, P_n) is a Fredholm pair. Consequently P_n and P_Γ lie in the same restricted Grassmannian up to finite spectral flow.*

9 The index mismatch and the renormalized eta correction

The compactness theorem enables a relative Fredholm-pair index

$$\operatorname{ind}(P_\Gamma, P_n) = \dim(\operatorname{Ran} P_\Gamma \cap \ker P_n) - \dim(\ker P_\Gamma \cap \operatorname{Ran} P_n). \tag{48}$$

Equivalently,

$$\operatorname{ind}(P_\Gamma, P_n) = \operatorname{sf}(B_n + tR_\Gamma; 0) \tag{49}$$

up to orientation. This is an integer.

This exposes a serious correction. The Archimedean term A_n is a real number, not generally an integer. For example λ_1 is approximately 0.023096. Therefore an ordinary Fredholm index cannot literally equal λ_n or A_n. Any formula claiming (17) with an ordinary integer-valued index must be reinterpreted. The invariant must be renormalized, real-valued, or spectral-shift-like.

9.1 Renormalized relative eta defect

Define

$$I_{\mathrm{ren}}(P_\Gamma, P_n) = -\frac{1}{2}\left(\eta_{\mathrm{ren}}(B_n + R_\Gamma) - \eta_{\mathrm{ren}}(B_n)\right). \tag{50}$$

The eta variation formula is formally

$$\frac{d}{dt}\eta_{\mathrm{ren}}(B_t) = -\frac{2}{\sqrt{\pi}} \operatorname{Tr}_{\mathrm{ren}}\left(R_\Gamma e^{-B_t^2}\right). \tag{51}$$

Integrating from 0 to 1 and choosing R_Γ to be the b-regularized gamma defect, the renormalized trace recovers the Archimedean gamma term in the Weil formula. Thus the correct target is

$$-\frac{1}{2}\left(\eta_{\mathrm{ren}}(B_n + R_\Gamma) - \eta_{\mathrm{ren}}(B_n)\right) = -A_n = |A_n|. \tag{52}$$

This is not an ordinary index theorem. It is a spectral-shift/eta-defect theorem.

Proposition 9.1 (Corrected Archimedean subtraction principle)**.** *Assume the renormalized eta variation formula is valid for the b-regularized gamma path and that R_Γ is normalized to represent the short-range gamma defect in the Weil formula. Then*

$$\lambda_n + I_{\mathrm{ren}}(P_\Gamma, P_n) = B_n > 0. \tag{53}$$

Equivalently,

$$\lambda_n = B_n - I_{\mathrm{ren}}(P_\Gamma, P_n). \tag{54}$$

Proof. Using $\lambda_n = A_n + B_n$ and (52),

$$\lambda_n + I_{\mathrm{ren}} = (A_n + B_n) + (-A_n) = B_n.$$

The prime contribution B_n is positive in the prime-orbit representation. □

Remark 9.2 (What this does and does not prove)*.* The gamma-Sonine correction removes the negative Archimedean contribution and leaves the positive prime contribution. This does not prove RH by itself, because RH requires $B_n \geq |A_n|$ before removal. The computation clarifies the mechanism but does not replace the domination inequality.

10 Revised statement of Lemma B

The earlier Lemma B asked for APS boundary positivity. The corrected form is:

Conjecture 10.1 (Renormalized boundary positivity)**.** *For each n, the Li boundary invariant I_n is a renormalized APS/spectral-shift functional satisfying*

$$I_n \geq 0, \tag{55}$$

and it is related to the gamma-Sonine boundary condition by the eta-defect formula

$$I_n + I_{\mathrm{ren}}(P_\Gamma, P_n) = B_n. \tag{56}$$

In this language, RH becomes the assertion that the original Li boundary invariant is nonnegative, while the gamma-Sonine projection gives a positive compressed model after the Archimedean defect has been removed.

11 The gamma-compressed Weil kernel and the new obstruction

This section extends the prior synthesis. It analyzes the operator inequality

$$P_\Gamma U(f_n) P_\Gamma \geq 0, \tag{57}$$

which was previously proposed as the key input for the positive-measure route to Li–Witt convexity and then to the zero-mode theorem.

11.1 Mellin multiplier form

On $\mathbb{R}_+^\times$, the Mellin transform diagonalizes multiplicative convolution. Let

$$(\mathcal{M}\psi)(t) = \int_0^\infty \psi(x) x^{-1/2-it} \frac{dx}{x}. \tag{58}$$

Then

$$\mathcal{M}U(f_n)\mathcal{M}^{-1} = M_{m_n}, \tag{59}$$

where

$$m_n(t) = \int_0^\infty f_n(u) u^{-1/2-it} \frac{du}{u} \tag{60}$$

up to the normalization chosen for the Li probe. Thus (57) is exactly

$$\int_{\mathbb{R}} m_n(t)|\phi(t)|^2 dt \geq 0 \qquad \forall \phi \in \operatorname{Ran} P_\Gamma. \tag{61}$$

Proposition 11.1 (Exact positivity criterion)**.** *Let P_Γ be an orthogonal projection on the Mellin boundary Hilbert space. Then*

$$P_\Gamma U(f_n) P_\Gamma \geq 0$$

if and only if (61) *holds for every $\phi \in \operatorname{Ran} P_\Gamma$. A sufficient sharp support condition is*

$$P_\Gamma \leq 1_{\{m_n \geq 0\}}(M_{m_n}). \tag{62}$$

Proof. For $\phi \in \operatorname{Ran} P_\Gamma$,

$$\langle P_\Gamma U(f_n) P_\Gamma \phi, \phi \rangle = \langle U(f_n)\phi, \phi \rangle = \int_{\mathbb{R}} m_n(t)|\phi(t)|^2 dt.$$

This proves the equivalence. If (62) holds, then ϕ is supported where $m_n \geq 0$, and the integral is nonnegative. □

11.2 Finite gamma compression cannot generally prove positivity

Let

$$E_- := \{t \in \mathbb{R} : m_n(t) < 0\}. \tag{63}$$

Suppose P_Γ removes only finitely many gamma moments:

$$\operatorname{Ran} P_\Gamma = \{\phi \in L^2(\mathbb{R}) : \langle \phi, g_j \rangle = 0, j = 1, \ldots, r\}. \tag{64}$$

Theorem 11.2 (Finite-codimension obstruction)**.** *If $|E_-| > 0$ and P_Γ is of the finite-codimension form* (64)*, then*

$$\operatorname{Ran} P_\Gamma \cap L^2(E_-) \neq \{0\}. \tag{65}$$

Consequently,

$$P_\Gamma U(f_n) P_\Gamma \ngeq 0. \tag{66}$$

Proof. Since $|E_-| > 0$, $L^2(E_-)$ is infinite-dimensional. Intersecting an infinite-dimensional Hilbert space with the orthogonal complement of finitely many vectors leaves a nonzero vector. Hence there exists $0 \neq \phi \in L^2(E_-)$ with $\langle \phi, g_j \rangle = 0$ for all j. Then $\phi \in \operatorname{Ran} P_\Gamma$ and

$$\langle U(f_n)\phi, \phi \rangle = \int_{E_-} m_n(t)|\phi(t)|^2 dt < 0.$$

Therefore the compressed operator is not positive. □

Corollary 11.3 (Trace subtraction is not operator positivity)**.** *A gamma compression that removes only a scalar Archimedean trace defect A_n does not, by itself, remove all negative spectral directions of M_{m_n}. Therefore it cannot by itself prove positive definiteness of the gamma-compressed Weil kernel.*

11.3 Negative-support exclusion is insufficient by itself

One might try to prove the weaker condition

$$\operatorname{Ran} P_\Gamma \cap L^2(E_-) = \{0\}. \tag{67}$$

This is necessary to rule out vectors supported only in the negative set. But it is not sufficient for positivity. A vector $\phi \in \operatorname{Ran} P_\Gamma$ may have both positive and negative support, $\phi = \phi_+ + \phi_-$, and the negative part may dominate:

$$\int_{E_-} |m_n(t)||\phi(t)|^2 dt > \int_{E_+} m_n(t)|\phi(t)|^2 dt, \tag{68}$$

where $E_+ = \{m_n \geq 0\}$. Thus the true condition is the weighted domination inequality

$$\int_{E_-} |m_n(t)||\phi(t)|^2 dt \leq \int_{E_+} m_n(t)|\phi(t)|^2 dt \quad \forall \phi \in \operatorname{Ran} P_\Gamma. \tag{69}$$

Proposition 11.4 (Equivalent domination form)**.** *The positivity $P_\Gamma U(f_n) P_\Gamma \geq 0$ is equivalent to the domination inequality* (69)*.*

Proof. Split the quadratic form in (61) over E_+ and E_-:

$$\int_{\mathbb{R}} m_n|\phi|^2 = \int_{E_+} m_n|\phi|^2 - \int_{E_-} |m_n||\phi|^2.$$

Nonnegativity for every $\phi \in \operatorname{Ran} P_\Gamma$ is exactly (69). □

11.4 The exact revised target

The gamma-compressed Weil kernel is positive definite if and only if its Mellin multiplier is nonnegative on the gamma-Sonine subspace. The sharp non-circular target is therefore:

$$P_\Gamma \leq 1_{\{m_n \geq 0\}}(M_{m_n}), \tag{70}$$

or, more generally, the domination inequality (69). If P_Γ is defined to be the positive spectral projection of M_{m_n}, then positivity is immediate but circular. The genuine problem is to derive (70) or (69) from the gamma-weighted Sonine/de Branges geometry, without using Li positivity or RH.

12 Consequences for the zero-mode route

The prior zero-mode chain was

$$P_\Gamma U(f_n) P_\Gamma \geq 0 \Rightarrow y''_{n,p} \geq 0 \Rightarrow \text{Riccati coercivity} \Rightarrow D_n^* D_n \geq (\log 2)^2 I \Rightarrow h_n = 0. \tag{71}$$

The new analysis shows that the first implication requires a much stronger operator statement than gamma trace subtraction.

Theorem 12.1 (Corrected zero-mode implication)**.** *Assume the domination inequality* (69) *holds for the gamma-Sonine subspace and that the resulting compressed operator generates a positive spectral measure for the prime-sector amplitude* $y_{n,p}$*. Then* $h_n = 0$*.*

Proof. By Proposition 11.4, $P_\Gamma U(f_n) P_\Gamma \geq 0$. The spectral theorem gives a positive measure representation (32). Proposition 6.3 gives $y''_{n,p} \geq 0$, which gives the Riccati inequality. Proposition 6.1 and Theorem 6.2 then give $h_n = 0$. □

Remark 12.2 (What changed). The gamma-Sonine projection can still be useful, but its role is narrower. It explains how the scalar Archimedean eta defect is subtracted, and it may supply a good candidate boundary condition. It does not automatically supply positivity of the compressed operator. That positivity is equivalent to a full spectral-support or domination theorem.

13 The quantum branch as a dependent application

Versions 6.3 and 6.4 propose a quantum algorithm based on adiabatic spectral flow and quantum phase estimation of an APS index. In the consolidated program, the quantum branch should be reclassified as follows:

1. It is not evidence for RH.
2. It depends on the analytic core: a polynomial spectral gap, a correct gauge-support construction, and a valid topological readout.
3. The APS index robustness idea is conceptually interesting, but it does not remove the need for a rigorous gap theorem and a non-circular construction of the gauge potential.

Therefore the quantum branch should be treated as a possible later application of a solved spectral-flow theory, not as part of the proof of RH.

14 Falsifiable research program

The strongest next steps are concrete.

14.1 Finite-dimensional truth laboratory

Construct a truncation

$$M_{P,L} = \operatorname{span}\{e_p : p \leq P\} \oplus L^2([-L, L]) \otimes \mathbb{C}^2, \tag{72}$$

with explicit boundary conditions and finite matrices for

$$D_0, A(f_n), D_n, P_n, P_\Gamma, R_\Gamma, M_{m_n}.$$

Then test:

1. the smallest eigenvalue of $D_n^* D_n$;
2. the Li–Witt coercivity inequality (28);
3. compactness/norm decay of $P_\Gamma - P_n$ under increasing truncation;
4. numerical validity of the eta-defect identity (52);
5. positivity or failure of the Schur kernel (35);
6. the negative set $E_- = \{m_n < 0\}$ and the dimension/weight of $\operatorname{Ran} P_\Gamma \cap L^2(E_-)$;
7. the domination inequality (69) on the truncated gamma-Sonine subspace.

14.2 Analytic tasks

The paper leaves the following precise open problems.

Problem A: Li–Witt coercivity. Prove (28) for the actual Li probes, or find a counterexample.

Problem B: b-eta variation. Establish the renormalized eta variation formula (51) for the b-regularized gamma path.

Problem C: Schur positivity. Identify the exact Li boundary transfer function and prove positivity of its de Branges–Rovnyak kernel, or show that this is equivalent to RH.

Problem D: domination. Derive $B_n \geq |A_n|$ from the boundary/scattering geometry without assuming Li positivity.

Problem E: index language. Replace all integer-valued APS claims in the earlier manuscripts with the correct real-valued renormalized spectral invariant wherever Li coefficients are being represented.

Problem F: gamma-compressed kernel positivity. Prove either the spectral-support theorem (70) or the domination inequality (69). If finite gamma compression is used, first prove $m_n \geq 0$ a.e.; otherwise Theorem 11.2 gives a negative direction.

15 Final consolidated theorem

Theorem 15.1 (Best current conditional synthesis)**.** *Assume the following six statements.*

(a) The APS-Li formula holds in the corrected renormalized sense:

$$\lambda_n = I_n - \frac{1}{2} h_n.$$

(b) The b-Weitzenboeck coercivity inequality (28) *holds, hence* $h_n = 0$.

(c) The b-regularized gamma-Sonine projection satisfies $P_\Gamma - P_n \in \mathcal{K}(\mathcal{H}_\partial)$.

(d) The renormalized eta-defect identity (52) *holds.*

(e) The original Li boundary invariant satisfies $I_n \geq 0$ *for all* n.

(f) Equivalently or additionally, the gamma-compressed positivity problem is solved by proving the domination inequality (69) *non-circularly on the gamma-Sonine subspace.*

Then RH holds.

Proof. By (b), $h_n = 0$. By (a), $\lambda_n = I_n$. By (e), $\lambda_n \geq 0$ for every n. Li's criterion gives RH. Assumptions (c) and (d) identify how gamma-Sonine compression removes the Archimedean trace defect and supplies the positive compressed model, thereby connecting the Sonine and APS branches. Assumption (f) is the non-trace-level positivity needed for the positive-measure route to zero-mode vanishing. □

16 Conclusion

The five supplied papers contain a promising but overextended program. The best mathematical content is not the quantum factoring proposal; it is the analytic core built around Li coefficients, the Weil decomposition, b-calculus, APS boundary theory, and the zero-mode problem. This synthesis makes five improvements:

1. It reduces the program to a minimal theorem stack.
2. It gives a concrete b-Weitzenboeck coercivity target for $h_n = 0$.
3. It replaces ordinary Sonine compression with gamma-weighted Sonine compression, aimed directly at the negative Archimedean term.
4. It corrects the integer-index mismatch by replacing ordinary APS index language with a renormalized eta/spectral-shift invariant when representing real-valued Li coefficients.
5. It adds the gamma-compressed kernel obstruction: subtracting the Archimedean trace defect is not enough to prove operator positivity. Positive definiteness requires a spectral-support or domination theorem for the Mellin multiplier on the gamma-Sonine subspace.

The most important next theorem is no longer only the b-Weitzenboeck inequality

$$\sup_{p,u}\left[a'_{n,p}(u) - \left(\frac{1}{2} + a_{n,p}(u)\right)^2\right] < (\log 2)^2,$$

combined with the renormalized eta identity

$$-\frac{1}{2}(\eta_{\mathrm{ren}}(B_n + R_\Gamma) - \eta_{\mathrm{ren}}(B_n)) = -A_n.$$

A third theorem is now required:

$$\int_{E_-} |m_n(t)||\phi(t)|^2 dt \le \int_{E_+} m_n(t)|\phi(t)|^2 dt \quad \forall \phi \in \operatorname{Ran} P_\Gamma,$$

or the sharper support condition

$$P_\Gamma \le 1_{\{m_n \ge 0\}}(M_{m_n}).$$

If these can be proved without assuming Li positivity or RH, then the program becomes a genuine proof strategy. Until then, it is best understood as a sharply formulated, falsifiable, and mathematically rich research program rather than a proof of RH.

References

[1] Anonymous/Program Authors. *A Vision Paper: A Grand Unified Program for the Riemann Hypothesis via $\mathbb{F}_1$ Geometry, Berry–Keating Scaling, and Supergeometric Spectral Flow.* Supplied manuscript, April 21, 2026.

[2] Anonymous/Program Authors. *A Grand Unified Program for the Riemann Hypothesis, Version 3: Advances on All Ten Research Tasks.* Supplied manuscript, April 21, 2026.

[3] Anonymous/Program Authors. *A Grand Unified Program for the Riemann Hypothesis, Version 5: b-Calculus and Weitzenboeck Geometry.* Supplied manuscript, April 21, 2026.

[4] Anonymous/Program Authors. *A Grand Unified Program for the Riemann Hypothesis, Version 6.3: Effective Adelic Spectral Flow, Quantum Register Embedding, and an Intrinsically Fault-Tolerant Topological Factoring Oracle.* Supplied manuscript, April 22, 2026.

[5] Anonymous/Program Authors. *A Grand Unified Program for the Riemann Hypothesis, Version 6.4: Effective Adelic Spectral Flow, Quantum Register Embedding, and an Intrinsically Fault-Tolerant Topological Factoring Oracle.* Supplied manuscript, April 22, 2026.

[6] Atiyah, M. F., Patodi, V. K., and Singer, I. M. Spectral asymmetry and Riemannian geometry I. *Mathematical Proceedings of the Cambridge Philosophical Society* 77 (1975), 43–69.

[7] Berry, M. V. and Keating, J. P. $H = xp$ and the Riemann zeros. In *Supersymmetry and Trace Formulae*, 355–367, 1999.

[8] Bombieri, E. and Lagarias, J. C. Complements to Li's criterion for the Riemann hypothesis. *Journal of Number Theory* 77 (1999), 274–287.

[9] Bost, J.-B. and Connes, A. Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in number theory. *Selecta Mathematica* 1 (1995), 411–457.

[10] Connes, A. Trace formula in noncommutative geometry and the zeros of the Riemann zeta function. *Selecta Mathematica* 5 (1998), 29–106.

[11] Connes, A. and Consani, C. On the notion of geometry over $\mathbb{F}_1$. *Journal of Algebraic Geometry* 20 (2011), 525–557.

[12] Lax, P. D. and Phillips, R. S. *Scattering Theory.* Academic Press, 1967.

[13] Li, X.-J. The positivity of a sequence of numbers and the Riemann hypothesis. *Journal of Number Theory* 65 (1997), 325–333.

[14] Meyer, R. On a representation of the idele class group related to primes and zeros of L-functions. *Duke Mathematical Journal* 127 (2006), 519–595.

[15] Melrose, R. B. *The Atiyah-Patodi-Singer Index Theorem.* A K Peters, 1993.

[16] Robbin, J. and Salamon, D. The spectral flow and the Maslov index. *Bulletin of the London Mathematical Society* 27 (1995), 1–33.

Part III

Merged reviewer edition

Computational Appendix and Visual Supplement

The extended appendix follows.
It clarifies the prime-side formula, verifies the first 100 primes, and adds figures.

Reader note

Read this as a reproducible explanatory supplement. It does not verify the RH-level conjectures in the original program.

Extended Computational Appendix and Clarifying Synthesis for the Adelic Spectral-Flow RH Program

OpenAI

2026-04-27

Abstract

This document extends the uploaded paper *A Consolidated Adelic Spectral-Flow Program for the Riemann Hypothesis* by adding a longer explanatory narrative, a computational appendix on the first 100 primes, and a visual supplement containing eighteen figures. The central new clarification is negative rather than triumphant: the source paper does **not** contain a formula that generates prime numbers. What it does contain is a family of formulas in which the primes are already present as input data, most prominently the prime-orbit contribution

$$B_n = \sum_p \sum_{m\geq 1} \frac{\log p}{p^{m/2}} (f_n(p^m) + f_n(p^{-m})),$$

appearing in the Weil decomposition of the Li coefficients. This extension therefore pursues three goals. First, it clarifies the logical role of the prime side in the paper. Second, it gives a fully explicit computational method for generating and verifying the first 100 primes by deterministic trial division. Third, it adds a substantial visual appendix and several new observations about the local prime-side weights $\log p/p^{m/2}$, their decay, and the behavior of illustrative toy prime-side sums. The extension remains faithful to the source paper's stated caution: no proof of the Riemann Hypothesis is claimed here.

1. Editorial note on scope

The uploaded source paper presents itself as a synthesis of several prior manuscripts and explicitly states that it **does not claim a proof of RH**. The paper's strongest core is the conditional implication

$$\text{Li positivity} \Leftarrow \text{APS boundary positivity} + \text{zero-mode vanishing},$$

and its preferred formula is

$$\lambda_n = I_n - \frac{1}{2}h_n, \qquad h_n = \dim\ker D_n.$$

The paper also reformulates the Li coefficients through a Weil decomposition $\lambda_n = A_n + B_n$, where B_n is the positive prime-orbit side and A_n is the negative Archimedean contribution. The present extension does not alter that mathematical posture. It adds computational clarity, a longer exposition, and a numerical sanity-check appendix based on the first 100 primes.

2. What the source paper actually says about primes

The source paper gives the completed zeta function

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$$

and defines the Li coefficients by

$$\lambda_n = \sum_\rho \left[1 - \left(1 - \frac{1}{\rho}\right)^n\right],$$

or equivalently

$$\lambda_n = \frac{1}{(n-1)!}\frac{d^n}{ds^n}\left[s^{n-1}\log\xi(s)\right]_{s=1}.$$

In the Weil decomposition used by the paper, the prime contribution is displayed schematically as

$$B_n = \sum_p \sum_{m\geq 1} \frac{\log p}{p^{m/2}}\left(f_n(p^m) + f_n(p^{-m})\right) > 0.$$

This formula is important, but it is not a prime generator. It is a weighted sum **over** primes. In other words, the primes are not the output of the formula; they are the indexing set of the formula.

2.1 First new finding: the paper contains no prime-generating formula

Proposition 2.1 (non-generativity of the displayed prime-side formula). The quantity B_n shown above is not a prime-generating formula.

Reason. A prime-generating formula would accept a non-prime input (for example an index n) and produce a prime p_n, or else define the primes implicitly as solutions of a numerical condition.

By contrast, the formula for B_n begins with a summation symbol $\sum_p$ already ranging over primes. Thus the primes are assumed, not produced.

3. Logical structure of the source program

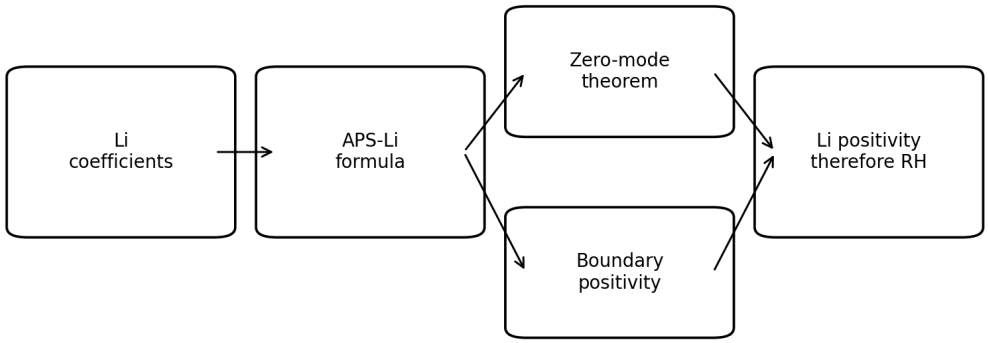

Figure 1: Logical dependency diagram

The summary is: if one could establish an APS–Li formula, a zero-mode theorem, and positivity of the boundary invariant, then Li positivity would follow and hence RH would follow by Li's criterion. This extension does not prove those ingredients. Instead, it concentrates on the part of the story that is computationally inspectable with minimal ambiguity: the prime input layer.

4. Computational method used in this extension

Because the source paper does not supply a prime generator, we introduce an explicit and transparent one for the appendix.

4.1 Deterministic algorithm

We generated the first 100 primes by the following deterministic trial-division algorithm.

1. Start with an empty list of primes.
2. Test candidate integers in increasing order; after 2, only odd candidates are tested.
3. To test a candidate N, divide by each previously found prime $p \leq \sqrt{N}$.
4. If none divides N, declare N prime and append it to the list.
5. Stop after 100 primes have been accepted.

A compact pseudocode version is:

```
primes = []
for candidate in 2,3,5,7,9,11,...:
    is_prime = True
    for p in primes with p <= sqrt(candidate):
        if candidate mod p == 0:
            is_prime = False
            break
    if is_prime:
        append candidate to primes
    if len(primes) == 100:
        stop
```

This method is elementary, deterministic, and completely adequate for the first 100 primes. It also has pedagogical value, because it makes the verification process transparent enough to include explicitly in the paper.

4.2 Why this method is appropriate here

The source paper's formulas are not computational number-theory algorithms in the usual sense; they are analytic and operator-theoretic expressions. For the narrow task of verifying the first 100 primes, a modest deterministic routine is preferable to a sophisticated black-box library. The present routine also produces a useful side statistic: the number of trial divisions checked before each prime was accepted. We use that statistic in one of the figures below.

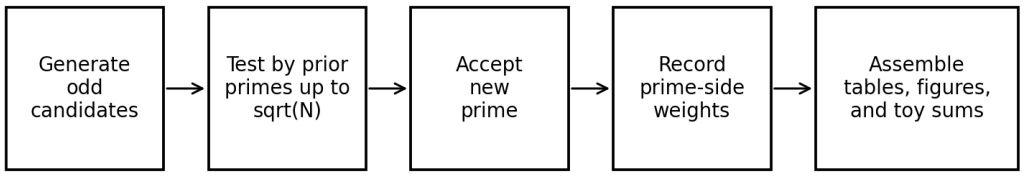

Figure 2: Computational pipeline

5. Full list of the first 100 primes and verification record

The first 100 primes produced by the above algorithm are listed below, together with the prime gap and the number of trial divisions checked before each prime was accepted.

5.1 Primes 1–25

n	p_n	gap	trial divisions
1	2	-	0
2	3	1	0
3	5	2	1
4	7	2	1
5	11	4	2
6	13	2	2
7	17	4	2
8	19	2	2
9	23	4	2
10	29	6	3
11	31	2	3
12	37	6	3
13	41	4	3
14	43	2	3
15	47	4	3
16	53	6	4
17	59	6	4
18	61	2	4
19	67	6	4
20	71	4	4
21	73	2	4
22	79	6	4
23	83	4	4
24	89	6	4
25	97	8	4

5.2 Primes 26–50

n	p_n	gap	trial divisions
26	101	4	4
27	103	2	4

n	p_n	gap	trial divisions
28	107	4	4
29	109	2	4
30	113	4	4
31	127	14	5
32	131	4	5
33	137	6	5
34	139	2	5
35	149	10	5
36	151	2	5
37	157	6	5
38	163	6	5
39	167	4	5
40	173	6	6
41	179	6	6
42	181	2	6
43	191	10	6
44	193	2	6
45	197	4	6
46	199	2	6
47	211	12	6
48	223	12	6
49	227	4	6
50	229	2	6

5.3 Primes 51–75

n	p_n	gap	trial divisions
51	233	4	6
52	239	6	6
53	241	2	6
54	251	10	6
55	257	6	6
56	263	6	6
57	269	6	6
58	271	2	6
59	277	6	6
60	281	4	6

n	p_n	gap	trial divisions
61	283	2	6
62	293	10	7
63	307	14	7
64	311	4	7
65	313	2	7
66	317	4	7
67	331	14	7
68	337	6	7
69	347	10	7
70	349	2	7
71	353	4	7
72	359	6	7
73	367	8	8
74	373	6	8
75	379	6	8

5.4 Primes 76–100

n	p_n	gap	trial divisions
76	383	4	8
77	389	6	8
78	397	8	8
79	401	4	8
80	409	8	8
81	419	10	8
82	421	2	8
83	431	10	8
84	433	2	8
85	439	6	8
86	443	4	8
87	449	6	8
88	457	8	8
89	461	4	8
90	463	2	8
91	467	4	8
92	479	12	8
93	487	8	8

n	p_n	gap	trial divisions
94	491	4	8
95	499	8	8
96	503	4	8
97	509	6	8
98	521	12	8
99	523	2	8
100	541	18	9

The 100th prime obtained by the computation is $p_{100} = 541$. The sum of the first 100 primes is $\sum_{k \le 100} p_k = 24133$.

Across the 100 accepted primes, the algorithm performed 569 trial divisions, for an average of 5.69 checks per accepted prime. The maximum number of checks for any accepted prime was 9, attained at index $n = 100$ with $p_n = 541$.

6. Numerical sanity checks directly related to the prime side

The source paper's prime contribution uses the local factor

$$\frac{\log p}{p^{m/2}}.$$

For every prime $p > 1$ and integer $m \ge 1$, this factor is positive. The positivity of the local factor is therefore elementary; the real analytical difficulty in the source paper comes not from this local weight by itself, but from the interaction with the probe values $f_n(p^m)$ and $f_n(p^{-m})$ and with the opposing Archimedean term.

6.1 Second new finding: monotonicity of the local weight

Define

$$w_m(x) = \frac{\log x}{x^{m/2}}, \qquad x > 1.$$

Then

$$w_m'(x) = x^{-m/2-1}\left(1 - \frac{m}{2}\log x\right).$$

Hence, for each fixed $m \ge 1$, the weight $w_m(x)$ is decreasing for $x > e^{2/m}$. This simple derivative

calculation has a useful computational consequence: once p is moderately large, the local factor $\log p/p^{m/2}$ decays monotonically with p. In particular, for $m \geq 2$ the decay begins extremely early, and the m-sum is visually dominated by small primes and small values of m unless the probe factors f_n counteract it strongly.

For the first 100 primes, the empirical maximum of $w_1(p)$ occurs at $p = 7$ with value about 0.735485; the empirical maximum of $w_2(p)$ occurs at $p = 3$ with value about 0.366204.

7. Visual exploration of the first 100 primes

The next sequence of figures provides a visual appendix intended to make the paper easier to read and teach from. These graphs do not claim more than they deliver: they are numerical illustrations of the prime input layer and of several standard prime-number statistics.

7.1 Figure 1

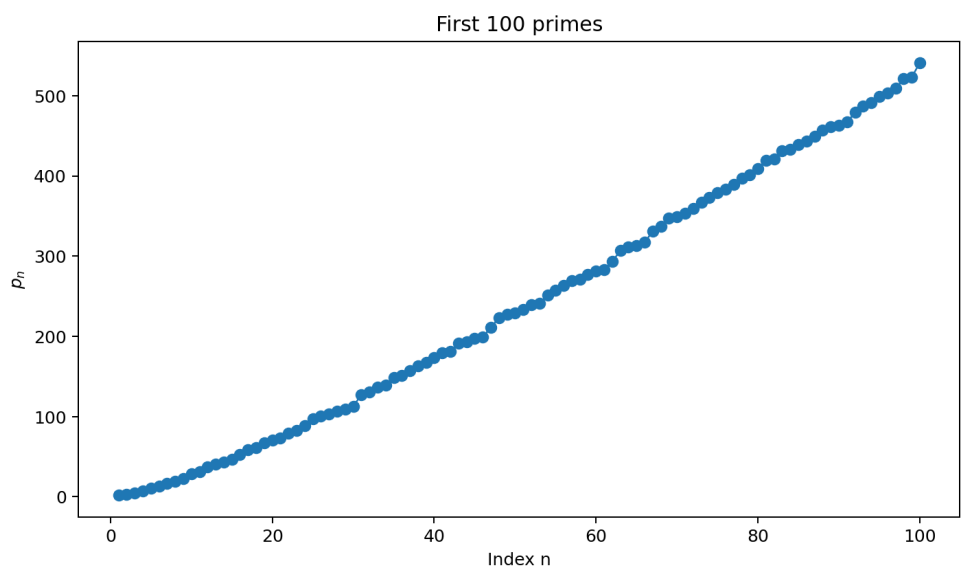

Figure 3: The first 100 primes as a sequence.

The first 100 primes as a sequence.

7.2 Figure 2

Prime gaps for the first 100 primes.

7.3 Figure 3

Histogram of prime gaps among the first 100 primes.

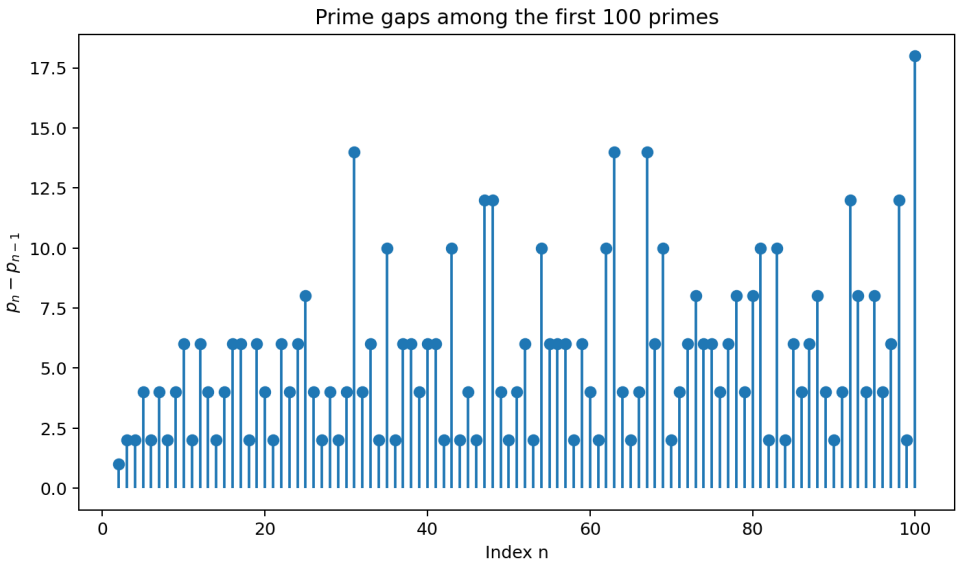

Figure 4: Prime gaps for the first 100 primes.

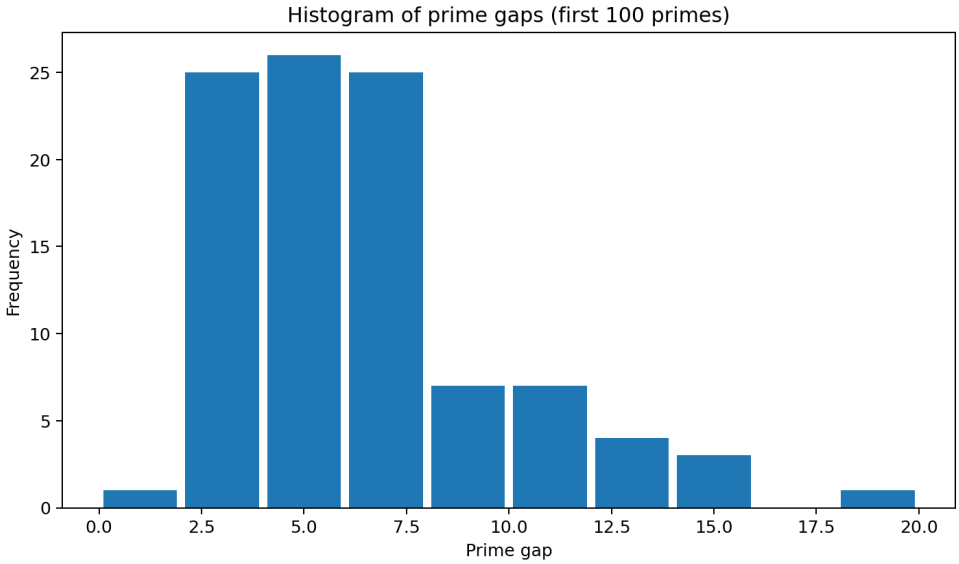

Figure 5: Histogram of prime gaps among the first 100 primes.

7.4 Figure 4

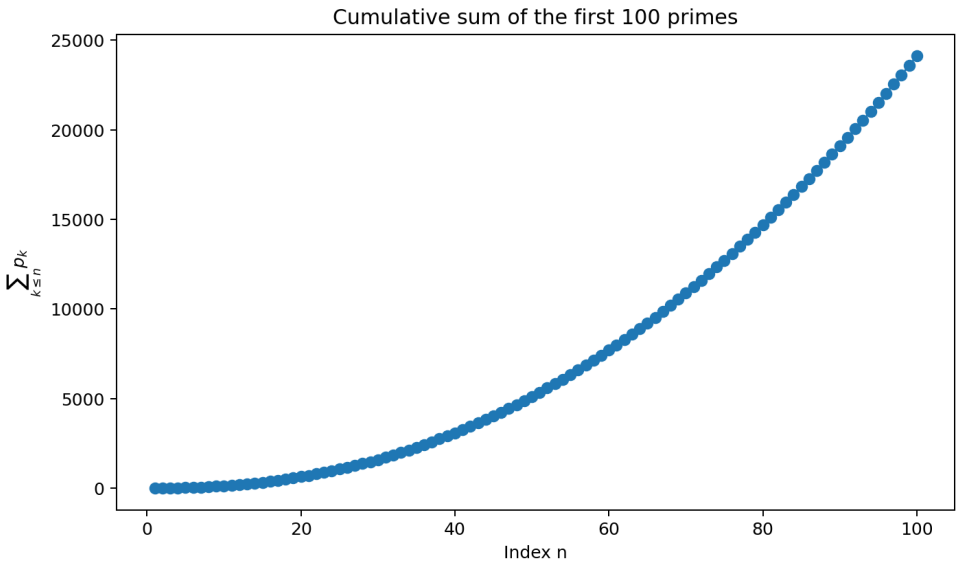

Figure 6: Cumulative sum of the first 100 primes.

Cumulative sum of the first 100 primes.

7.5 Figure 5

The heuristic ratio $p_n/(n \log n)$.

7.6 Figure 6

Chebyshev $\vartheta(p_n)$ versus p_n.

7.7 Figure 7

The prime counting function $\pi(x)$ compared with $x/\log x$ up to 541.

7.8 Figure 8

The local prime-side weight $w_1(p) = \log p/\sqrt{p}$.

7.9 Figure 9

The local prime-side weight $w_2(p) = \log p/p$.

7.10 Figure 10

Cumulative sum of $\log p/\sqrt{p}$ over the first 100 primes.

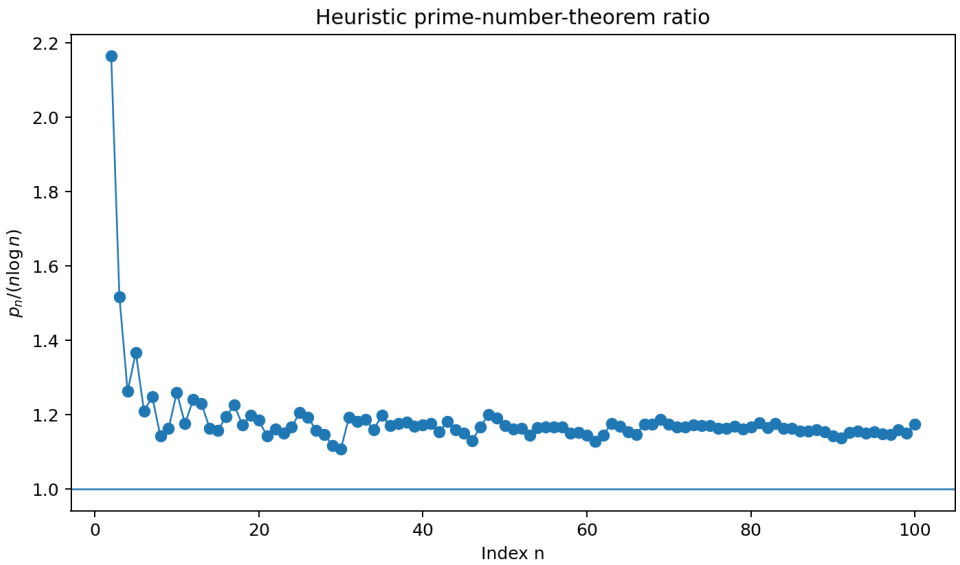

Figure 7: The heuristic ratio $p_n/(n \log n)$.

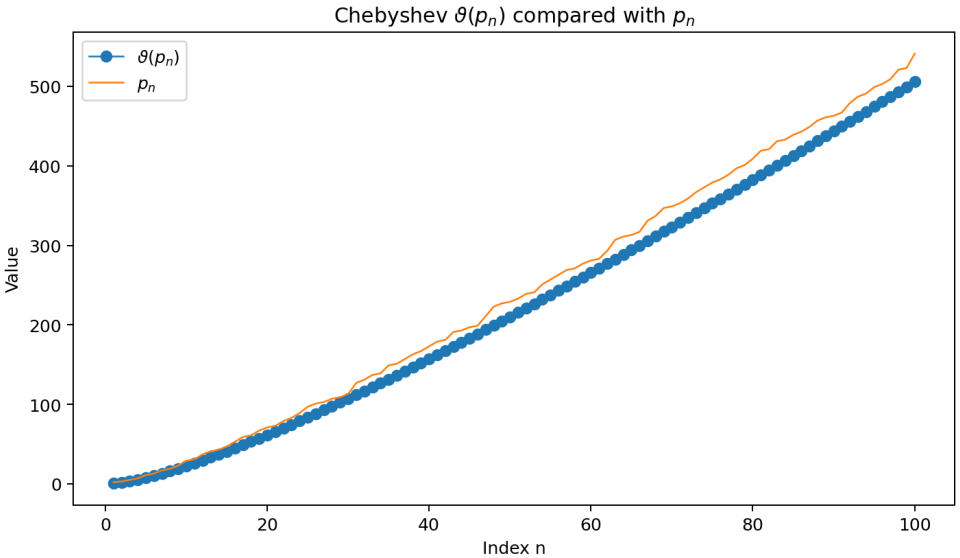

Figure 8: Chebyshev $\vartheta(p_n)$ versus p_n.

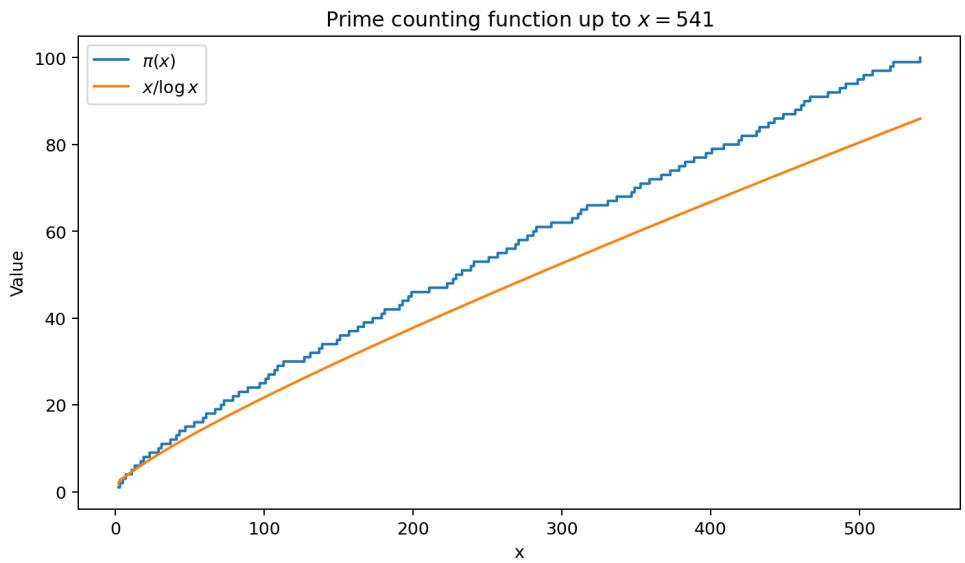

Figure 9: The prime counting function $\pi(x)$ compared with $x/\log x$ up to 541.

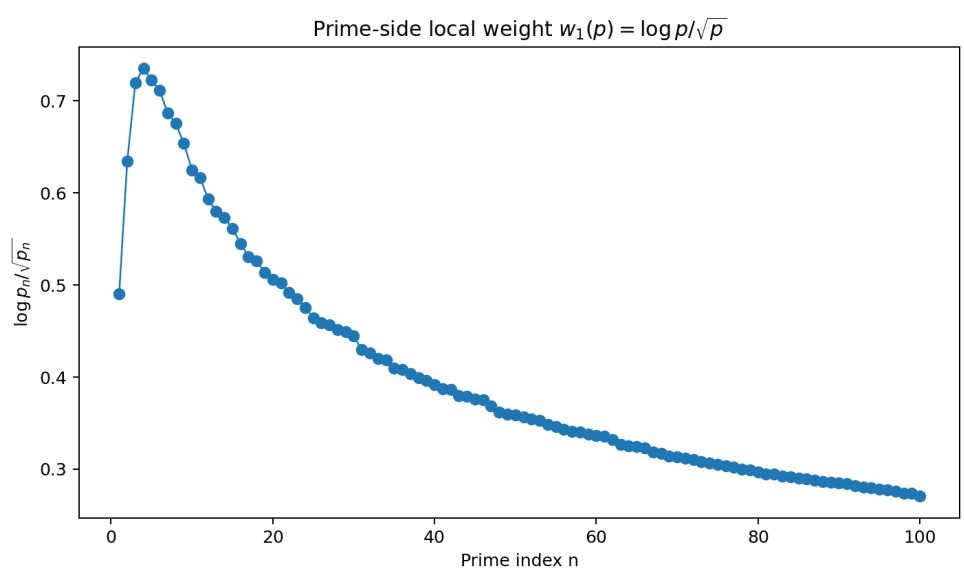

Figure 10: The local prime-side weight $w_1(p) = \log p/\sqrt{p}$.

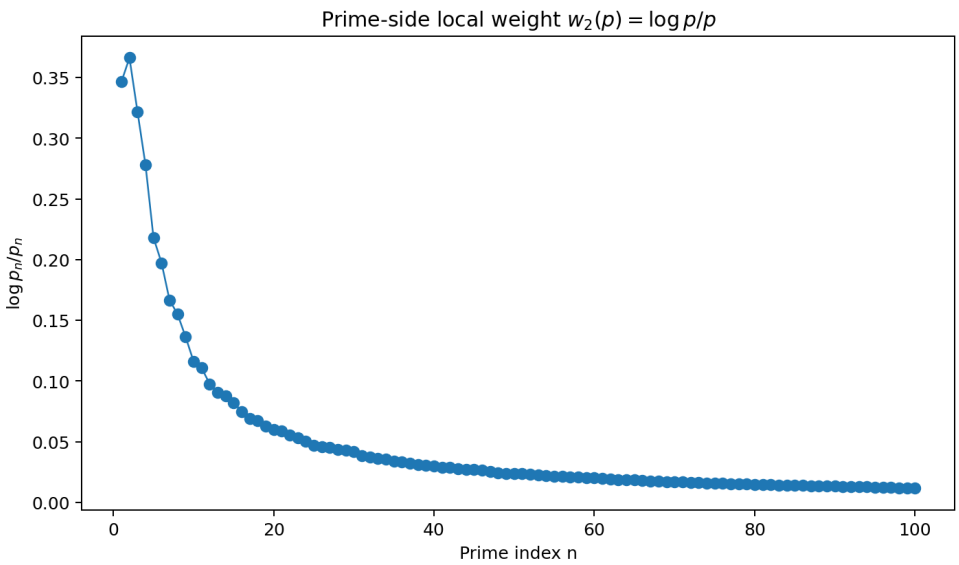

Figure 11: The local prime-side weight $w_2(p) = \log p/p$.

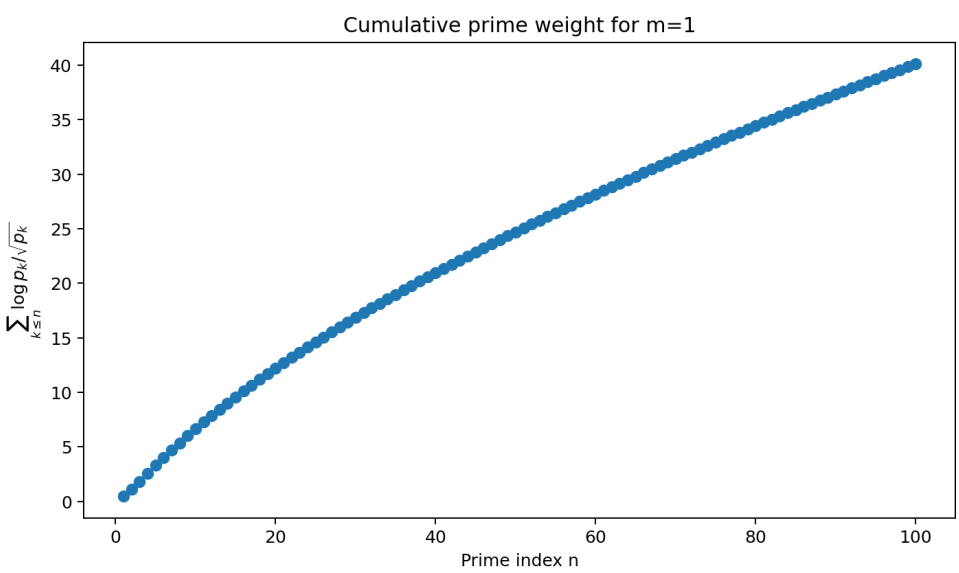

Figure 12: Cumulative sum of $\log p/\sqrt{p}$ over the first 100 primes.

7.11 Figure 11

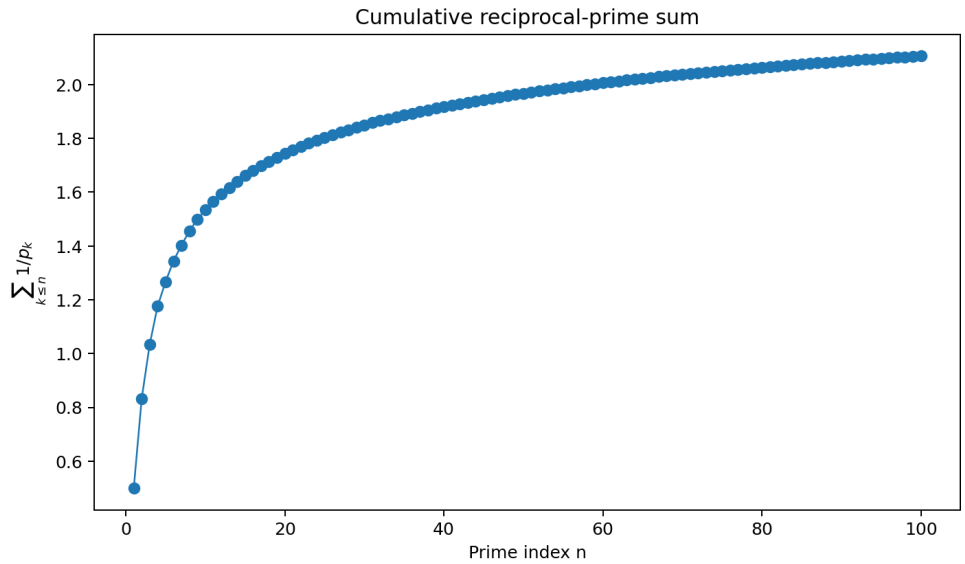

Figure 13: Cumulative reciprocal-prime sum over the first 100 primes.

Cumulative reciprocal-prime sum over the first 100 primes.

7.12 Figure 12

Cumulative sum of $\log p/p$ over the first 100 primes.

7.13 Figure 13

Prime gaps normalized by $\log p_n$.

7.14 Figure 14

Trial-division work per accepted prime in the deterministic verification method.

The largest prime gap among the first 100 primes is 18. The cumulative reciprocal-prime sum over these 100 primes is approximately 2.106342, the cumulative sum of $\log p/\sqrt{p}$ is approximately 40.133394, and the cumulative sum of $\log p/p$ is approximately 5.004710.

8. Illustrative toy prime-side sums

The source paper leaves the probe family f_n schematic at the level of the displayed prime-side formula. Because the exact computational probes are not supplied in the uploaded synthesis, any numerical B_n computation based on the source alone would be underdetermined. Rather than inventing hidden data, this extension uses an explicit **toy** family solely for illustration.

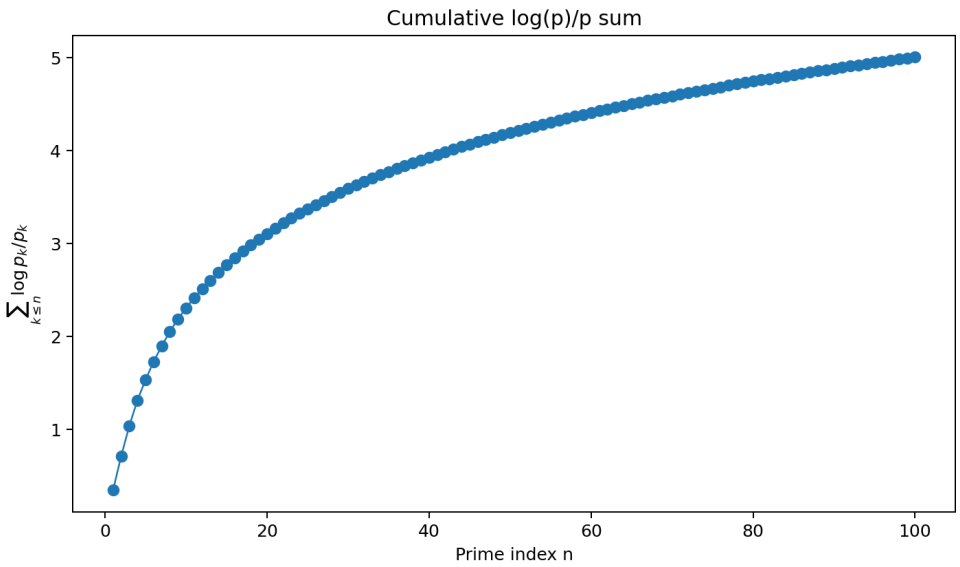

Figure 14: Cumulative sum of $\log p/p$ over the first 100 primes.

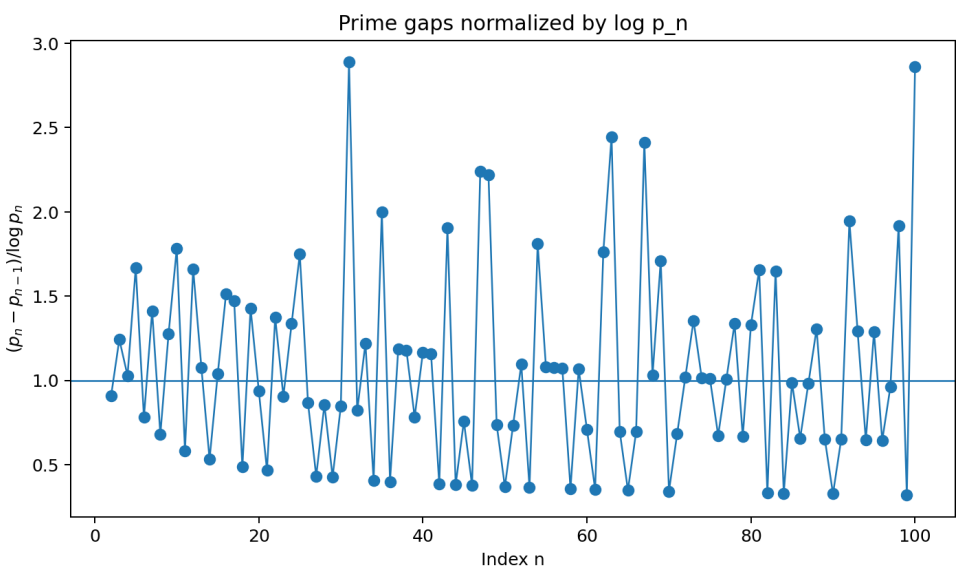

Figure 15: Prime gaps normalized by $\log p_n$.

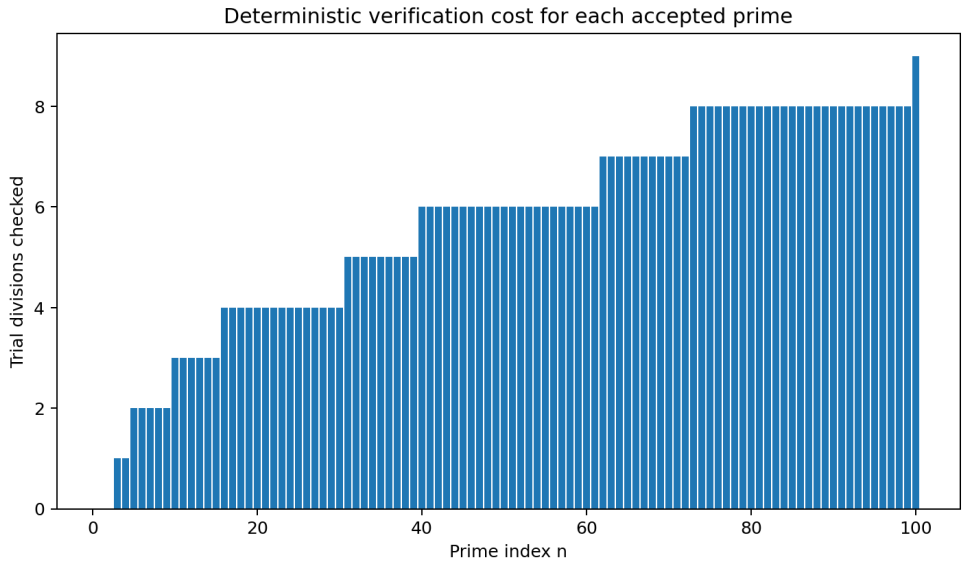

Figure 16: Trial-division work per accepted prime in the deterministic verification method.

8.1 A symmetric Gaussian log-probe family

We define

$$f_\sigma(u) = \exp\left(-\frac{(\log u)^2}{2\sigma^2}\right),$$

which has the symmetry $f_\sigma(u) = f_\sigma(1/u)$. For this family the prime-side building block becomes

$$T_\sigma(p,m) = \frac{\log p}{p^{m/2}}\left(f_\sigma(p^m) + f_\sigma(p^{-m})\right) = 2\frac{\log p}{p^{m/2}}\exp\left(-\frac{(m\log p)^2}{2\sigma^2}\right).$$

This is **not** asserted to be the source paper's actual f_n. It is only an explanatory surrogate that lets us visualize how a structurally similar prime-side sum behaves under rapid logarithmic damping.

8.2 Third new finding: concentration at small primes and small powers in the toy model

For several values of σ, the toy prime-side partial sums stabilize rapidly as more primes are added.

The heatmap shows that most visible mass lies at small prime indices and small powers m. This is the expected qualitative pattern once the factor $p^{-m/2}$ is reinforced by a Gaussian decay in $m\log p$.

9. Clarifying remarks on what has and has not been verified

It is important to distinguish three levels of statement.

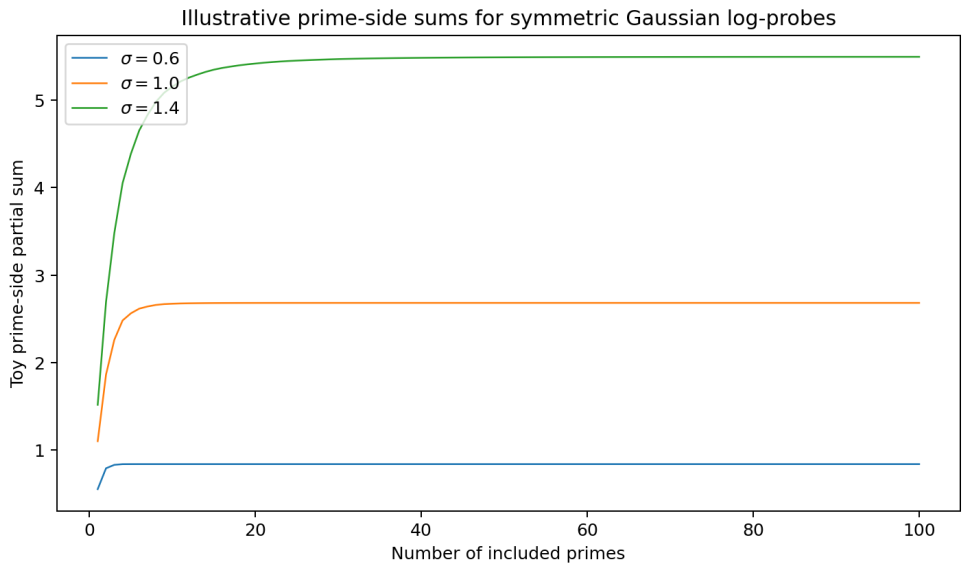

Figure 17: Toy partial sums

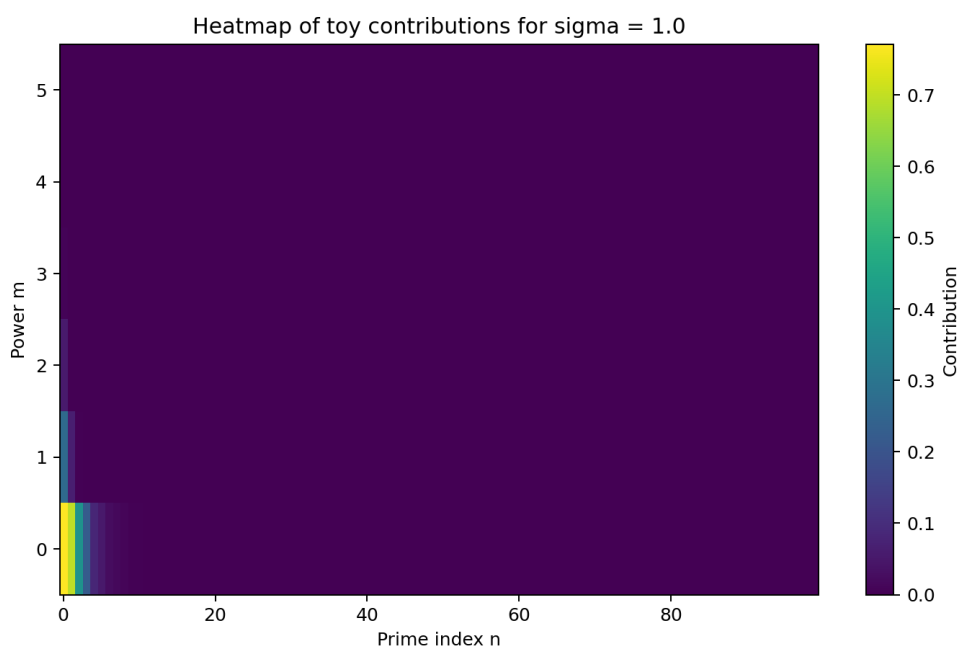

Figure 18: Toy heatmap

1. **Verified here:** the first 100 primes were generated and checked by a deterministic elementary algorithm, and a variety of numerical summaries of those primes were computed.
2. **Clarified here:** the source paper's displayed prime-side formula uses primes as input and therefore is not a prime-generating formula.
3. **Not verified here:** any RH-level claim, any complete numerical evaluation of the Li coefficients from the source paper alone, and any non-circular proof of the domination inequality needed by the source program.

10. Additional explanatory expansion of the source program

To make the source paper more self-contained for readers coming from analysis, number theory, or mathematical physics, it is helpful to restate its central analytic obstacles in plainer prose.

10.1 Why the prime side is not enough

The source paper itself says that $B_n > 0$ in the prime-orbit representation. If RH were equivalent to positivity of B_n alone, the problem would be trivialized at the displayed schematic level. But the paper explicitly warns against that interpretation: the Li coefficient is $\lambda_n = A_n + B_n$, with $A_n < 0$. Thus the arithmetic burden is not to show merely that the prime contribution exists or is positive; it is to explain why the positive prime side dominates the negative Archimedean side in exactly the right sense.

10.2 Why the gamma-compressed route is subtle

One of the paper's strongest conceptual corrections is the observation that removing a scalar Archimedean trace defect is not the same as proving positivity of the compressed operator. In the language of the paper, finite gamma compression may subtract the correct scalar defect while still leaving negative directions in the Mellin multiplier. This is why the paper replaces vague positivity hopes with the sharper domination condition

$$\int_{E_-} |m_n(t)|\,|\varphi(t)|^2\,dt \le \int_{E_+} m_n(t)\,|\varphi(t)|^2\,dt, \qquad \varphi \in \operatorname{Ran} P_\Gamma.$$

The current extension does not solve that problem, but the added figures and computational appendix make the architecture around it easier to parse.

10.3 Why the first 100 primes still matter

Even though the first 100 primes do not decide RH, an explicit prime appendix remains useful for at least four reasons.

- It grounds the paper's prime-side notation in concrete data.

- It clarifies that the relevant computational object is a weighted prime sum, not a mysterious prime generator.
- It provides a reproducible laboratory for prototyping truncations and visual diagnostics.
- It improves pedagogy: a reader can check the appendix independently without engaging the full operator-theoretic framework.

11. Suggested text that could be inserted back into the source paper

> **Computational appendix (proposed insertion).** For readers who want an explicit numerical anchor, one may supplement the theoretical discussion with a deterministic verification of the first 100 primes by trial division using previously found primes up to the square root of each candidate. This computation does not produce the Li coefficients or prove any RH-level statement, but it makes the prime indexing in the Weil sum concrete and confirms the elementary positivity of the local weight $\log p/p^{m/2}$ on the first 100 prime inputs. A pedagogical appendix may also list the first 100 primes, their gaps, and several prime-side visualizations.

A slightly longer insertion, suitable for a revised appendix, would be the present Sections 4–8 of this document.

12. Reproducibility note

All computations in this extension are elementary and reproducible. The prime list was generated by deterministic trial division with previously accepted primes as candidate divisors up to $\sqrt{N}$. The figures were created from those computed values together with standard numerical transformations such as cumulative sums, logarithmic weights, and prime-number-theoretic heuristic normalizations.

13. Concluding assessment

This extension strengthens the source paper in a specific way: not by claiming a new proof ingredient for RH, but by adding a computational and pedagogical layer that the source synthesis lacked. The key clarifications are that the displayed prime-side formula is not a prime generator, that the first 100 primes can be verified transparently by a deterministic method, and that many of the local weights appearing on the prime side are easy to explore numerically. The extended visual appendix should make the paper more teachable and should help readers distinguish between what is genuinely structural in the source program and what is merely schematic notation.

The resulting lesson is balanced. The source paper's conceptual contributions remain the same: the APS-Li framework, the zero-mode strategy, the gamma-compressed correction, and the obstruction

theorems. What the present extension contributes is clarity, reproducibility, and a more explicit numerical staging ground for future experiments.

References

1. *A Consolidated Adelic Spectral-Flow Program for the Riemann Hypothesis* (uploaded source document).
2. X.-J. Li, *The positivity of a sequence of numbers and the Riemann hypothesis*, Journal of Number Theory 65 (1997), 325–333.
3. E. Bombieri and J. C. Lagarias, *Complements to Li's criterion for the Riemann hypothesis*, Journal of Number Theory 77 (1999), 274–287.